CISS240: Introduction to Programming
Quiz q1704

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This is a closed-book, no C++ compiler, 6-minute quiz.

The following questions relate to bubblesorting the following array of integers:

```
5   1   8   3   7   9
```

Q1. How many comparisons are carried out in the first pass? (i.e. how many times the $<$ or $\leq$ or $>$ or $\geq$ comparison was executed in the first pass.)
Answer:

```
5
```

Q2. How many swaps are carried out in the first pass?
Answer:

```
3
```

Q3. What are the values in the array at the end of the first pass? Write it in this form: `{1, 2, 3, 4}`. This is not the right answer – it's just to provide a format for the answer.
Answer:

```
{1, 5, 3, 7, 8, 9}
```

Q4. What are the values in the array at the end of the second pass? Write it in this form: `{1, 2, 3, 4}`. This is not the right answer – it's just to provide a format for the answer.
Answer:

```
{1, 3, 5, 7, 8, 9}
```

The following are the values in a sorted array of integers:

```
1   3   5   7   8   9
```

The following are questions related to the binary search algorithm. The index variables are `lower`, `upper`, and `mid`. Of course `lower` is initialized to 0 and `upper` is

initialized to 5 and during the search, the first value assigned to `mid` is 2.

Q5. If the binary search is executed with the aim to find the value 7, during the first iteration, which variable (`lower` or `upper`) is changed?
Answer:

```
lower
```

Q6. Continuing the above question, what is the new value of the variable that was changed in the above question?
Answer:

```
3
```

Q7. Write down all the values of `mid` during the binary search for `1` in the above array. Write it in this form: `1 2 3 4`. This is not the right answer – it's just to provide a format for the answer.
Answer:

```
2 0
```

Q8. Write down all the values of `mid` during the binary search for `6` in the above array. Write it in this form: `1 2 3 4`. This is not the right answer – it's just to provide a format for the answer.
Answer:

```
2 4 3
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.